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PHY 101: General Physics I - Mechanics, Matter and Waves

Build the physical intuition and problem-solving habits that General Physics rewards

FACULTY PRIMER

Faculty Primer

Faculty of Physical Sciences
Physics & Industrial Physics | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

This pack treats General Physics as a small number of powerful ideas applied repeatedly. You get a structured method for solving mechanics problems, a clear treatment of simple harmonic motion and waves, and a formula sheet with the conditions under which each formula is valid.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Solve kinematics and dynamics problems using a consistent sign convention and free-body diagrams.
- Apply conservation of energy and momentum to collisions and connected bodies.
- Explain simple harmonic motion and calculate period, frequency and energy.
- Describe wave properties including superposition, interference and standing waves.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

Problem-solving protocol

Physics marks are lost in the setup, not in the algebra. Use the same five steps every time until they are automatic.

1. Draw a diagram and label known and unknown quantities with units.
2. Choose a coordinate system and state which direction is positive.
3. Write the relevant principle - Newton's second law, energy conservation, or momentum conservation.
4. Substitute symbols first, then numbers. This makes errors easier to spot and earns method marks.
5. Check magnitude, units and whether the direction of the answer is physically sensible.

> **Free-body diagrams are not optional**
> If two bodies are in contact or connected by a string, draw each body separately. Most 'impossible' pulley problems become simple once the tensions and weights are separated.

Mechanics essentials

Kinematics describes motion; dynamics explains it. Keep them distinct in your answers.

- Equations of motion apply only when acceleration is constant.
- Newton's second law: net force equals mass times acceleration. Net means the vector sum.
- Friction: static friction can take any value up to the limiting value; kinetic friction is roughly constant.
- Work done equals force times displacement in the direction of the force.
- Conservation of mechanical energy applies only when no non-conservative forces do work.
- Momentum is conserved in collisions; kinetic energy is conserved only in elastic collisions.

Oscillations and waves

Simple harmonic motion appears in springs, pendulums, molecules and circuits. Learn the defining relationship and the rest follows.

- Defining condition: acceleration is proportional to displacement and directed toward equilibrium.
- Period of a mass-spring system: T = 2 pi times the square root of m/k.
- Period of a simple pendulum for small angles: T = 2 pi times the square root of L/g.
- Total energy in SHM is proportional to the square of the amplitude.
- Wave speed equals frequency times wavelength.
- Superposition explains interference, beats and standing waves; nodes are points of zero displacement.

> **Angle in radians**
> The small-angle approximation sin(theta) is approximately theta only when theta is in radians. Using degrees silently corrupts the pendulum formula.

Measurement, uncertainty and vectors

Physics begins with measurement and the admission that measurement has limits. Every reading has an uncertainty, and a result cannot be more precise than its inputs. Significant figures are a shorthand for that logic; unit consistency is a check that will catch most errors before they cost marks.

Vectors have magnitude and direction; scalars do not. Displacement, velocity, acceleration and force are vectors; distance, speed, mass and energy are scalars. Resolving a vector into components turns a two-dimensional problem into two one-dimensional problems, which is the single most useful technique in mechanics.

Kinematics describes motion without asking its cause: displacement, velocity, acceleration and time, related by the equations for constant acceleration. Dynamics asks why motion changes, and answers with force and mass.

Newton's laws and conservation

Newton's first law states that a body remains at rest or in uniform motion unless a net external force acts. The second law quantifies the relationship: net force equals rate of change of momentum, or F = ma for constant mass. The third law states that forces occur in equal and opposite pairs acting on different bodies - which is why they do not cancel.

Conservation laws are often easier than force analysis. Momentum is conserved in the absence of external force, which solves collisions and explosions quickly. Energy is conserved but transforms, and mechanical energy is conserved only where no non-conservative force does work, such as friction.

Friction, tension, normal reaction and weight must be drawn accurately on a free-body diagram. Most errors in mechanics problems are diagram errors, not algebra errors.

> **Draw the free-body diagram first**
> One object, every force acting on it, nothing else. If your diagram is wrong, no amount of correct algebra will save the answer.

Circular motion, gravitation and simple machines

Circular motion requires a centripetal force directed toward the centre; the object's speed may be constant while its velocity changes continuously. The common error is inventing an outward "centrifugal" force in an inertial frame - what is felt is inertia, not a real force.

Newton's law of gravitation states that the force between two masses is proportional to the product of the masses and inversely proportional to the square of the distance. From this follow orbital motion, satellite speed and Kepler's laws as consequences rather than separate facts.

Simple machines - lever, pulley, inclined plane, wheel and axle - trade force for distance. Efficiency is useful work out divided by work in, and it is always less than one hundred per cent because of friction. The principle of moments for a lever is a conservation argument in disguise.

WORKED EXAMPLES

Do it once, slowly

These are not model answers to copy. They show the reasoning a strong student would use, step by step, so you can repeat the process on a question you have not seen.

1. A projectile problem

A ball is thrown at 20 m/s at 30 degrees above the horizontal. Find the time of flight and range (take g = 10 m/s2).

1. Resolve velocity: horizontal 20 cos30 = 17.3 m/s; vertical 20 sin30 = 10 m/s.
2. Time to highest point: v = u - gt gives 0 = 10 - 10t, so t = 1 s.
3. Total flight time: 2 s, by symmetry.
4. Range: horizontal velocity times total time = 17.3 x 2 = 34.6 m.
5. Maximum height: u squared over 2g = 100 / 20 = 5 m.
6. Check units and reasonable magnitude before writing the answer.

> **What to carry forward**
> Resolve, treat axes independently, then recombine. Never mix horizontal and vertical.

2. A momentum problem

A 2 kg trolley moving at 3 m/s collides and sticks to a stationary 1 kg trolley. Find the final velocity.

1. State conservation of momentum: total momentum before equals after.
2. Before: 2 x 3 + 1 x 0 = 6 kg m/s.
3. After: (2 + 1) x v = 3v.
4. Equate: 3v = 6, so v = 2 m/s in the original direction.
5. Check energy: kinetic energy falls from 9 J to 6 J, lost as heat and deformation - consistent with an inelastic collision.

> **What to carry forward**
> Momentum is conserved in all collisions; kinetic energy only in elastic ones.

EXAM TECHNIQUE

Questions with model answers

Written in the shape your course is usually examined, with the marking logic spelled out. Attempt each one under time before you read the outline.

Q1. State Newton's three laws and explain why the third law does not produce equilibrium.

15 marks

1. State each law precisely.
2. Explain that action and reaction act on different bodies.
3. Give a worked example, such as a book on a table.
4. Distinguish the reaction pair from the normal force and weight pair.

> **Where marks are lost**
> Treating weight and normal reaction as an action-reaction pair. They act on the same body.

Q2. Derive an expression for the speed of a satellite in a circular orbit.

15 marks

1. Set gravitational force equal to centripetal force.
2. Simplify to v squared = GM/r.
3. Discuss the dependence on orbital radius.
4. Note the assumption of a circular orbit and negligible drag.

REFERENCE

Glossary

Use these definitions in your own sentences; a copied definition rarely earns full marks.

Vector. A quantity with both magnitude and direction.

Scalar. A quantity with magnitude only.

Displacement. Change in position, a vector quantity.

Acceleration. Rate of change of velocity.

Free-body diagram. A diagram showing all forces acting on a single object.

Momentum. The product of mass and velocity, conserved absent external force.

Impulse. The product of force and time, equal to change in momentum.

Centripetal force. The net inward force required for circular motion.

Work. Force multiplied by displacement in the direction of the force.

Conservation of energy. The principle that energy is transformed but not created or destroyed.

Efficiency. Useful output divided by total input, expressed as a percentage.

Uncertainty. The range within which a measured value is likely to lie.

Your four-week plan

Week 1 - Kinematics

Describing motion.

- Solve ten constant-acceleration problems.
- Draw motion graphs and interpret gradients.
- Resolve five vectors into components.

Week 2 - Dynamics

Forces.

- Draw free-body diagrams for six situations.
- Apply F = ma to two-body systems.
- Explain the third law with a counter-example.

Week 3 - Conservation

Momentum and energy.

- Solve five collision problems.
- Check whether kinetic energy is conserved.
- Apply the work-energy theorem.

Week 4 - Circular motion

Rotation and gravity.

- Derive orbital speed.
- Solve three centripetal force problems.
- Explain why "centrifugal force" is not a real force.

Where to read more

Entry-level texts you can find in the university library or a departmental reading list. Confirm editions with your lecturer.

- Any first-year mechanics textbook with worked examples and problem sets.
- A practical physics manual for measurement and uncertainty.
- A revision guide for projectile and momentum problems.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. A body starts from rest and accelerates at 2 m/s^2 for 5 s. Find its displacement.

Answer: s = ut + 0.5at^2 = 0 + 0.5 x 2 x 25 = 25 m.

2. State the condition for mechanical energy to be conserved.

Answer: Only conservative forces do work; no friction, air resistance or other non-conservative force removes energy from the system.

3. A spring of constant 200 N/m is stretched 0.1 m. Find the stored energy.

Answer: E = 0.5 k x^2 = 0.5 x 200 x 0.01 = 1 J.

4. A wave has frequency 50 Hz and wavelength 4 m. Find its speed.

Answer: v = f x wavelength = 50 x 4 = 200 m/s.

5. Why do we draw separate free-body diagrams for connected bodies?

Answer: Each body has its own forces and acceleration; separating them prevents incorrectly adding tensions and weights to the same object.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Write the four constant-acceleration equations and state when they apply.
- [ ] Solve ten problems using free-body diagrams, one per problem.
- [ ] Derive the period of a mass-spring system in your own steps.
- [ ] Draw a standing wave and label nodes and antinodes.
- [ ] Build a one-page formula sheet with validity conditions, not just formulas.
- [ ] Do one timed past-question paper and mark every lost mark by cause.
- [ ] Explain one mechanics topic aloud to a coursemate without notes.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date:

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